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1)

Aim: To design and implement a Python-based resume information extraction system using Regular Expressions to extract candidate details, technical skills, and years of experience, and to generate structured candidate profiles and shortlist eligible candidates with at least 2 years of experience and Python skills.

Program:

```
import re

# Sample resume data
resumes = [
    """
    Name: Rahul Sharma
    Email: rahul@gmail.com
    Mobile: 9876543210
    Skills: Python, SQL, Machine Learning
    Experience: 3 years
    """,
    """
    Name: Priya Reddy
    Email: priya@gmail.com
    Mobile: 9123456780
    Skills: Java, NLP
    Experience: 1 year
    """,
    """
    Name: Arun Kumar
    Email: arun@gmail.com
    Mobile: 9988776655
    Skills: Python, Java, NLP
    Experience: 2 years
    """
```

```
]

# Extract information from resume
def extract_resume(text):
    name = re.search(r"Name:\s*(.*)", text)
    email = re.search(r"[\w.-]+@[\w.-]+\.\w+", text)
    mobile = re.search(r"\b[6-9]\d{9}\b", text)
    experience = re.search(r"Experience:\s*(\d+)\s*years?", text,
re.I)

    # Detect technical skills
   skill_list = ["Python", "Java", "SQL",
                 "Machine Learning", "NLP"]

    skills = [
        skill for skill in skill_list
        if re.search(r"\b" + re.escape(skill) + r"\b",
                     text, re.I)
    ]

    years = int(experience.group(1)) if experience else 0

    profile = {
        "Name": name.group(1).strip() if name else "Not found",
        "Email": email.group() if email else "Not found",
        "Mobile": mobile.group() if mobile else "Not found",
        "Skills": skills,
        "Experience": years
    }

    return profile


# Process and display all candidates
print("CANDIDATE PROFILES")
print("-" * 40)

eligible_candidates = []
```

```
for resume in resumes:
    candidate = extract_resume(resume)

    print("\nName:", candidate["Name"])
    print("Email:", candidate["Email"])
    print("Mobile:", candidate["Mobile"])
    print("Skills:", ", ".join(candidate["Skills"]))
    print("Experience:", candidate["Experience"], "years")

    # Eligibility criteria
    if (candidate["Experience"] >= 2 and
            "Python" in candidate["Skills"]):
        eligible_candidates.append(candidate)

# Display shortlisted candidates
print("\nELIGIBLE CANDIDATES")
print("-" * 40)

for candidate in eligible_candidates:
    print(candidate["Name"], "-",
          candidate["Experience"], "years -",
          ", ".join(candidate["Skills"]))
```



2)

Aim: To develop a Python-based resume information extraction system using Regular Expressions to extract candidate details, technical skills, and years of experience, and to shortlist candidates with at least 2 years of experience and Python skills.

Program:

```
import re

resumes = [
    """Name: Rahul Sharma
    Email: rahul@gmail.com
    Mobile: 9876543210
    Skills: Python, SQL, Machine Learning
    Experience: 3 years""",

    """Name: Priya Reddy
    Email: priya@gmail.com
    Mobile: 9123456780
    Skills: Java, NLP
```

```
    Experience: 1 year"""
]

def extract_resume(text):
    name = re.search(r"Name:\s*(.*)", text)
    email = re.search(r"[\w.-]+@[\w.-]+\.\w+", text)
    mobile = re.search(r"\b[6-9]\d{9}\b", text)
    exp = re.search(r"Experience:\s*(\d+)", text)

    skill_list = ["Python", "Java", "SQL",
                  "Machine Learning", "NLP"]

    skills = [s for s in skill_list
              if re.search(r"\b" + re.escape(s) + r"\b",
                           text, re.I)]

    years = int(exp.group(1)) if exp else 0

    return {
        "Name": name.group(1).strip() if name else "Not found",
        "Email": email.group() if email else "Not found",
        "Mobile": mobile.group() if mobile else "Not found",
        "Skills": skills,
        "Experience": years
    }

print("CANDIDATE DETAILS")

for resume in resumes:
    c = extract_resume(resume)

    print("\nName:", c["Name"])
    print("Email:", c["Email"])
    print("Mobile:", c["Mobile"])
    print("Skills:", ", ".join(c["Skills"]))
    print("Experience:", c["Experience"], "years")

    if c["Experience"] >= 2 and "Python" in c["Skills"]:
        print("Status: Eligible")
```

```
    else:
        print("Status: Not Eligible")
```

```
CANDIDATE DETAILS

Name: Rahul Sharma
Email: rahul@gmail.com
Mobile: 9876543210
Skills: Python, SQL, Machine Learning
Experience: 3 years
Status: Eligible

Name: Priya Reddy
Email: priya@gmail.com
Mobile: 9123456780
Skills: Java, NLP
Experience: 1 years
Status: Not Eligible
```

3)

Aim: To design and implement a Python-based student registration validation system using Regular Expressions to validate the register number, institutional email, course code, semester, and mobile number, and to generate a final registration status report.

Program

```
import re

# Student details
reg_no = "123456789012"
email = "student@college.edu"
course = "AIDS101"
semester = "3"
mobile = "9876543210"

# Validation
checks = {
    "Register Number": bool(re.fullmatch(r"\d{12}", reg_no)),
    "Institutional Email": bool(re.fullmatch(
```

```
        r"[\w.-]+@[\w.-]+\.edu", email)),
    "Course Code": bool(re.fullmatch(
        r"[A-Z]{2,5}\d{3}", course)),
    "Semester": bool(re.fullmatch(r"[1-8]", semester)),
    "Mobile Number": bool(re.fullmatch(
        r"[6-9]\d{9}", mobile))
}

# Display validation report
print("STUDENT REGISTRATION REPORT")

for field, valid in checks.items():
    print(field + ":", "Valid" if valid else "Invalid")

if all(checks.values()):
    print("\nRegistration Successful")
else:
    print("\nRegistration Failed")
```

```
STUDENT REGISTRATION REPORT
Register Number: Valid
Institutional Email: Valid
Course Code: Valid
Semester: Valid
Mobile Number: Valid

Registration Successful
```